- Consider $f:(0,a)\to\mathbb{R}$, $g:(0,a)\to\mathbb{R}$.

Check overleaf for 1st portion

we write $f(h)=O(g(h))$, if $\exists\,\varepsilon, C>0$ s.t.,

$$|f(h)|\le C|g(h)|,\quad \forall\, 0<h<\varepsilon$$

we write $f(h)=o(g(h))$ if

$$\frac{|f(h)|}{|g(h)|}\to 0,\quad h\to 0 \quad ?$$

Example

$\alpha_n=\frac{1}{n}$, $\beta_n=\frac{2}{n}$, $\gamma_n=\frac{1}{n^2}$

$$\frac{\beta_n}{\alpha_n}\le 2 \Rightarrow \beta_n=O(\alpha_n)$$

$$\frac{\gamma_n}{\alpha_n}=\frac{1}{n}\to 0 \Rightarrow \gamma_n=o(\alpha_n)$$

Let (lin. convergence) suppose $\{x_n\}_{n=1}^{\infty}$ that converges to x^*, i.e.

$\lim_{n\to\infty} x_n = x^*$ if $\exists\,\rho\in[0,1]$, s.t.

$$\lim_{n\to\infty}\frac{|x_{n+1}-x^*|}{|x_n-x^*|}=\rho$$

then
- $\{x_n\}\to x^*$ linearly if $\rho\in[0,1)$
- $\{x_n\}\to x^*$ superlinearly if $\rho=0$
- $\{x_n\}\to x^*$ sublinearly if $\rho=1$.

$x_n = 1+2^{2-n}$

$x_{n+1}=2^{-n}x_n$

$x_n = 1/n$

#1 $x_n = 1 + 2^{2-n} \to 1$

check linearity

$$\frac{|x_{n+1} - x^*|}{|x_n - x^*|} \Rightarrow \frac{|1 + 2^{1-n} - 1|}{|1 + 2^{2-n} - 1|} = \frac{|2/2^n|}{|2^2/2^n|}$$

$$= 1/2 \in (0,1)$$

$\downarrow$ reals limit

$1/2$

$\Rightarrow$ linearly converges.

$$x_{n+1} = 2^{-n} x_n = \frac{x_n}{2^n} \to 0 = x^*$$

$$\frac{|2^{-n} x_n - 0|}{|x_n|} = \frac{1}{2^n} \to 0 = \rho$$

superlinear

#3 $x_n = 1/n \to 0$

$$\left| \frac{1/(n+1)}{1/n} \right| = \frac{n}{n+1} \to 1 \Rightarrow \rho = 1$$

$\therefore$ x_n converges to 0 sublinearly

def [convergence with order p]: If $\exists p \geq 1$, $C > 0$, s.t.

$$\lim_{n\to\infty} \frac{|x_{n+1} - x^*|}{|x_n - x^*|^p} = C$$

Then $\{x_n\}_{n=1}^{\infty} \to x^*$ with order p

Remark: if $\{x_n\} \to x^*$ with order p, it will converge with order q, $0 < q < p$.

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right), \quad n = 0, 1, 2, \ldots \quad \text{where } a > 0.$$

① Suppose $x_n \to x^*$ i.e. $\lim_{n\to\infty} x_n = x^*$

$$\Rightarrow \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} \frac{1}{2}\left(x_n + \frac{a}{x_n}\right)$$

$$\Rightarrow x^* = \frac{1}{2}\left(x^* + \frac{a}{x^*}\right) \Rightarrow \frac{1}{2}x^* = \frac{1}{2}\frac{a}{x^*}$$

$$\Rightarrow (x^*)^2 = a$$

$$\Rightarrow x^* = \sqrt{a}.$$

WTS $x_n \to x^*$ with order 2.

$$\frac{|x_{n+1} - x^*|}{|x_n - x^*|^2} = \frac{\left|\frac{1}{2}\left(x_n + \frac{a}{x_n}\right) - x^*\right|}{|x_n - x^*|^2}$$

cont.

$$= \frac{\frac{1}{2}\left(x_n + \frac{a}{x_n}\right) - \sqrt{a}}{(x_n - \sqrt{a})^2} = \frac{1}{2}\,\frac{\left(\sqrt{x_n} - \frac{\sqrt{a}}{\sqrt{x_n}}\right)^2}{(x_n - \sqrt{a})^2}$$

$$= \frac{1}{2x_n}\,\frac{(x_n - \sqrt{a})^2}{(x_n - \sqrt{a})^2} = \frac{1}{2x_n} \xrightarrow{n\to\infty} \frac{1}{2\sqrt{a}} = C$$

$\Rightarrow x_n \to \sqrt{a}$ with order of 2

Iterative Method $F(x) = 0$ find seq. $\{x_n\} \to x^*$ s.t. $F(x^*) = 0$.

Methods

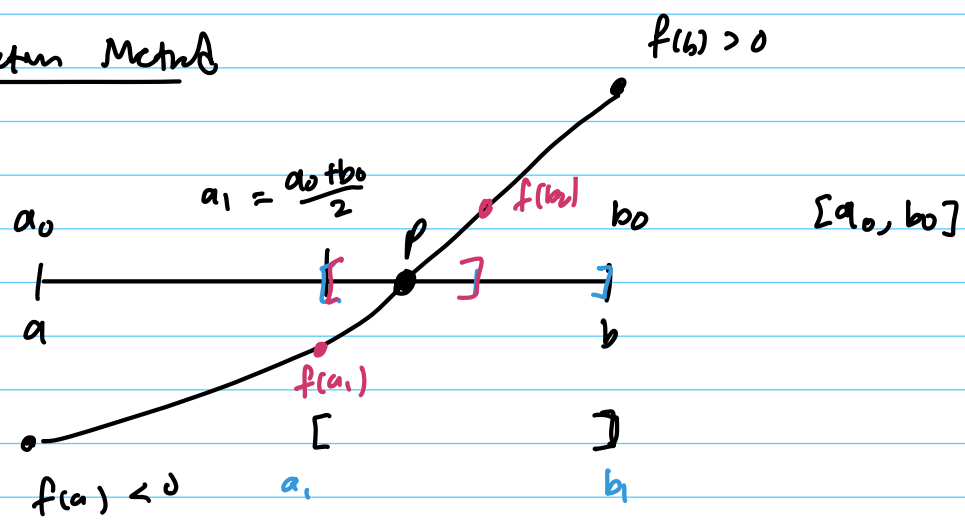
- Bisection Method
- Fixed-point iteration } Newton's Method.

<u>Thm</u> (IVT): If $f \in C[a,b]$, and k is between $f(a)$ and $f(b)$, then $\exists p \in [a,b]$ s.t. $f(p) = k$.

How do we find it?

<u>Corr</u>: If $f \in C[a,b]$ & $f(a)f(b) < 0$, $\exists p \in [a,b]$ s.t. $f(p) = 0$

<u>Bisection Method</u>



$[a_0, b_0]$

Note $[a_0, b_0] \supset [a_1, a_2] \supset [a_2, b_2] \supset \cdots$

$\bigcap_i [a_i, b_i] \neq \emptyset$ In fact $\lim_{n\to\infty} \bigcap_i [a_i, b_i] = p$

at each step either $a_{n+1} = a_n + \frac{b_n - a_n}{2}$

$$b_{n+1} = a_n + \frac{b_n - a_n}{2}.$$